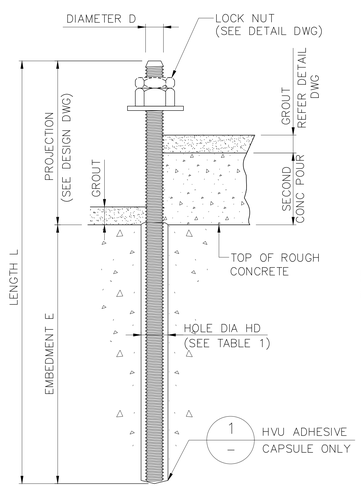
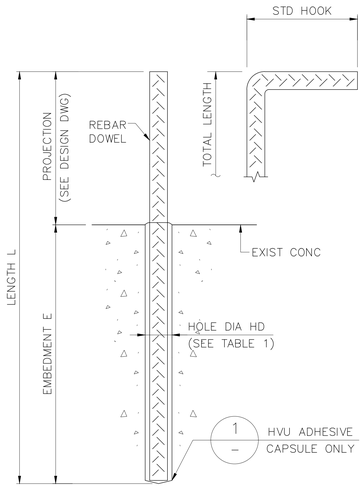


THREADED ROD

TYPE TR (CARBON STEEL)

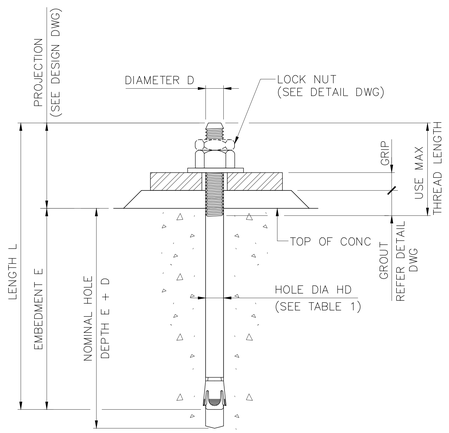
TYPE TRSS (STAINLESS STEEL)



REBAR DOWEL

TYPE RD (REBAR DOWEL)

TYPE RDH (REBAR DOWEL W/HOOK)



HILTI* KWIK BOLT 3

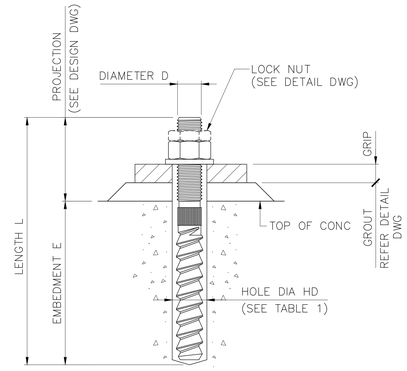
TYPE KB3 (CARBON STEEL)

TYPE KB3SS (STAINLESS STEEL)

HILTI* KWIK BOLT TZ

TYPE KBTZ (CARBON STEEL)

TYPE KBTZSS (STAINLESS STEEL)



HILTI* HIT-Z

TYPE Z (CARBON STEEL)

TYPE ZSS (STAINLESS STEEL)

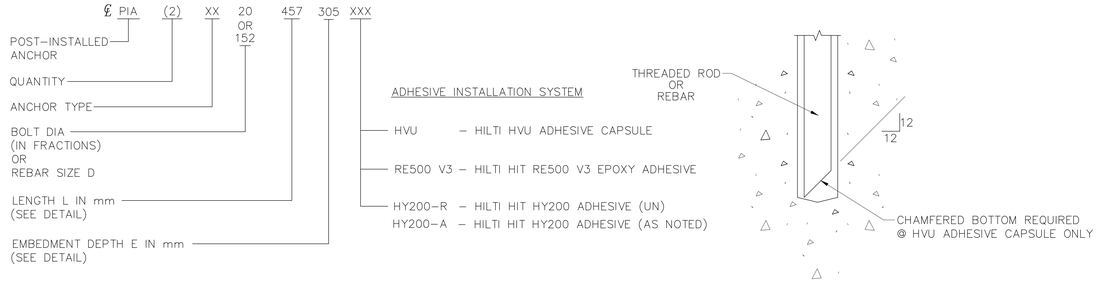
TABLE 1 (HOLE DIAMETERS)

NOMINAL DIA (D)	THREADED ROD HOLE DIAMETER (HD)			REBAR SIZE	REBAR DOWEL HOLE DIAMETER (HD)			KWIK BOLT 3 (HD)	KWIK BOLT TZ (HD)	HIT-Z HOLE DIA (HD)
	HVU	RE500 V3	HY200		HVU	RE500 V3	HY200	KB *	KB *	HY200
12	15	15	15	D10	–	12	12	12	12	15
16	18	20	18	D12	16	16	16	16	16	20
20	22	22	18	D16	21	20	20	20	20	22
22	24	24	24	D20	24	22	22	–	–	–
24	29	29	29	D22	29	24	24	24	–	–
29	32	32	–	D25	32	29	29	–	–	–
32	35	35	35	D28	–	35	35	–	–	–
35	–	–	–	D32	–	38	38	–	–	–
38	–	–	–	D35	–	45	–	–	–	–

* CONSULT HILTI TECHNICAL MANUAL FOR SIZE / MATERIAL AVAILABILITY

SPECIAL NOTES:

1. ALL DIMENSIONS ABOVE ARE IN mm
2. HILTI ANCHOR BOLTS OR EQUIVALENT TO BE USED
3. FOR ITEMS NOT SHOWN, HOLE DIAMETER SHALL BE AS NOTED ON THE DESIGN DRAWINGS
4. ANCHOR BOLTS SPACING & EDGE DISTANCE SHALL BE AS NOTED ON THE DESIGN DRAWINGS
5. FOR SHEAR & TENSION CAPACITY OF ANCHOR BOLTS, REFER TO HILTI MANUAL



POST-INSTALLED ANCHOR CALLOUT

DETAIL 1 / – (HVU ADHESIVE CAPSULE ONLY)

GENERAL NOTES:

1) FOR STRUCTURAL GENERAL NOTES SEE DRAWING MD-520-1000-EG-ST-STD-0009_A01.
2) ANCHORS SHALL NOT BE REPAIRED, REPLACED OR FIELD MODIFIED WITHOUT THE APPROVAL OF STRUCTURAL ENGINEER OF RECORD. INSTALLATION TOLERANCES SHALL CONFORM TO PIP STS03001 AND SECTION 7.5 OF CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIDGES, MARCH 2005.
3) FIELD SUBSTITUTION OF ANCHORS IN ANY FORM, WITHOUT WRITTEN AUTHORIZATION FROM THE ENGINEER OF RECORD, IS STRICTLY PROHIBITED.
4) INSTALLATION OF POST-INSTALLED ANCHORS INCLUDING DRILLING AND CLEANING OF HOLES AND INSTALLATION SHALL BE IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS.
5) MATERIAL FOR TYPE TR SHALL CONFORM TO ASTM F1554 GRADE36 (UNLESS OTHERWISE NOTED ON THE DESIGN DRAWINGS).
6) NUTS FOR TYPE TR SHALL BE IN ACCORDANCE WITH ANSI B18.2.2 HEAVY HEX (GRADE B) AND CONFORM TO ASTM A563/A563M, GRADE A.
7) THREADED ROD, NUTS AND WASHERS FOR TYPE TR SHALL BE HOT DIP GALVANIZED IN ACCORDANCE WITH ASTM A153M (UNLESS OTHERWISE NOTED ON THE DESIGN DRAWINGS).
8) MATERIAL FOR TYPE TRSS SHALL BE 304 STAINLESS STEEL IN ACCORDANCE WITH ASTM F1554 GRADE36 (UNLESS OTHERWISE NOTED ON THE DESIGN DRAWINGS).
9) NUTS FOR TYPE TRSS SHALL BE 304 STAINLESS STEEL IN ACCORDANCE WITH ASTM A563/A563M.
10) THREADS SHALL BE COARSE THREAD SERIES UNC FOR TYPES TR / TRSS ANCHORS 25mm IN DIAMETER OR LESS, AND 8 THREAD SERIES FOR TYPES TR / TRSS ANCHORS OVER 25mm IN DIAMETER. THE THREADS SHALL CONFORM TO ANSI B1.1 HAVING A CLASS 2A TOLERANCE FOR BOLTS, AND A CLASS 2B TOLERANCE FOR NUTS.
11) USE STANDARD CUT WASHERS WITH TYPE TR CONFORMING TO ASTM F436/F436M.WASHER MATERIAL FOR TYPE TRSS SHALL MATCH MATERIAL OF THREADED ROD.WASHER MATERIAL FOR TYPE TRSS SHALL MATCH MATERIAL OF THREADED ROD.
12) FOR TYPES TR / TRSS PROVIDE ONE (1) NUT AND ONE (1) WASHER UNLESS OTHERWISE NOTED ON DETAIL DRAWING. NUTS AND WASHERS FOR TYPE KB3, KBTZ AND TYPE Z SHALL BE AS SUPPLIED BY HILTI.
13) MATERIAL FOR TYPES RD / RDH (REBAR) SHALL BE DEFORMED BILLET STEEL AND CONFORM TO ASTM 615M OR ASTM A706M, GRADE 60.
14) MATERIAL FOR TYPES KB3 / KB3SS, KBTZ / KBTZSS AND Z SHALL BE IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS.
15) ANCHORS SHALL BE THOROUGHLY CLEANED OF RUST, THREAD CUTTING OIL, OR ANY OTHER SUBSTANCE WHICH WOULD REDUCE BOND, AND LENGTH.
16) VENDOR SHALL ATTACH A WEATHERPROOF TAG FOR IDENTIFICATION OF TYPES TR, TRSS, RD, AND RDH ANCHORS ACCORDING TO THEIR MARK NUMBER. TYPES KB3, KB3SS, KBTZ, KBTZSS, AND Z SHALL BE SHIPPED IN MANUFACTURER'S PACKAGING WHICH SHALL CLEARLY IDENTIFY SIZE AND LENGTH.
17) FOR INSTALLATION REQUIRING NO MANUAL HOLE CLEANING, COMBINE HILTI HIT HY200 SAFE SET ADHESIVE SYSTEM WITH HILTI HIT-Z (OR HIT-ZSS) ANCHORS.

REFERENCE SPECIFICATIONS:

1. MD-520-1000-EG-CV-SPC-0002 CONCRETE WORK SPECIFICATION
2. MD-520-1000-EG-CV-CRT-0001 CIVIL AND STRUCTURAL DESIGN CRITERIA – COMMON

Ma'aden Phosphate 3 – Phase 1

NAME	TITLE	SIGN	DATE

REVIEW GRADE		
Result	Code	Description
☐	A	Accepted
☐	C	Proceed with exceptions to next Submission and Revision
☐	R	Do Not Proceed - Revise as noted and Resubmit

Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due and proper performance of the Works in accordance with the Contract/Purchase Order.

PROJECT MANAGER SIGN-OFF:

NAME		DATE	

ISSUED FOR CONSTRUCTION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
			JKR	S.Suthar	D.Chauhan	B.Mistry	
C01	30-JUL-24	ISSUED FOR CONSTRUCTION	JKR	SPS	DMC	BDM	MA'ADEN
A03	26-OCT-23	ISSUED FOR APPROVAL	DAP	SPS	PSR	BDM	MA'ADEN
A02	22-SEP-23	ISSUED FOR APPROVAL	DAP	SPS	PSR	BDM	MA'ADEN
A01	28-AUG-23	ISSUED FOR REVIEW	DAP	SPS	PSR	BDM	MA'ADEN

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"Contractor Insignia to be placed here if applicable"

Worley energy | chemicals | resources

JESA INTERNATIONAL

MA'ADEN معادن Saudi Arabian Mining Company شركة معادن العربية السعودية

DRAWN	D.Patel	03-AUG-23
CHECKED	S.Suthar	15-AUG-23
ENGINEER	P.Rana	08-AUG-23
APPROVED	B.Mistry	28-AUG-23
EMPLOYER	MA'ADEN	–
SCALE	NTS	A1

Ma'aden Phosphate 3 Phase I
RAK & WAS
STRUCTURAL STANDARDS
POST-INSTALLED ANCHOR DETAILS

DRAWING NUMBER MD-520-1000-EG-CV-STD-0006 REV C01

$FILE$ $TIME$ $DATE$